

EpiKG: End-to-End Epistemic Preservation in Clinical Knowledge Graphs for Assertion-Aware Retrieval-Augmented Generation

Anonymous Author(s)

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Abstract

Clinical NLP systems can detect assertion status—negation, uncertainty, family history—with high accuracy, yet to our knowledge no existing system preserves this epistemic metadata through knowledge graph construction to retrieval-augmented generation. This *epistemic propagation gap* causes RAG systems to conflate mentioned conditions with present ones, retrieving “patient denies diabetes” as evidence of diabetes. We present EpiKG, a clinical knowledge graph system that preserves assertion status end-to-end through a 7-value assertion schema, a tri-temporal edge model with Allen’s interval algebra, shared concept nodes with cross-patient assertion statistics, and an assertion-aware GraphRAG pipeline. We introduce two benchmarks on MIMIC-IV data: ClinicalBench (400 assertion-sensitive questions across 9 epistemic categories) and Slice Bench (144 longitudinal questions across 3 patient complexity tiers). On ClinicalBench, vanilla RAG *degrades* LLM accuracy by 9.0 percentage points (71.5%→62.5%), while epistemic KG-RAG recovers to baseline accuracy with higher clinical safety scores (0.775 vs. 0.752) and dramatic gains on uncertainty (+20 pp) and family-history (+70 pp) categories. On Slice Bench, the KG contribution concentrates in high-complexity patients: +5.0 pp for patients with 15+ encounters versus +0.6 pp for 1–2 encounters, an eightfold difference. These results demonstrate that structured epistemic context is most valuable precisely when document volume overwhelms simple retrieval, and that assertion-blind knowledge graphs can be worse than no graph at all.

1 Introduction

A physician documents “patient denies chest pain, sister had MI at 45, will consider statin if lipids remain elevated”—a single sentence encoding four epistemic states: a negated symptom, a family-history event, a hypothetical action, and an implicit present condition. Clinical NLP can detect these assertions accurately [1, 2], yet when a RAG system ingests the same note, assertion context is flattened: negation is lost, family attribution collapses onto the patient, and downstream reasoning may conclude the patient *has* chest pain and *had* an MI. We call this the *epistemic propagation gap*—the systematic loss of assertion status as clinical information moves from extraction through storage to retrieval.

The gap persists because, to our knowledge, no system connects assertion detection to knowledge representation. The i2b2 2010 six-class taxonomy [1] remains the standard, yet we are unaware of a subsequent system that propagates these labels beyond extraction. OMOP excludes negated conditions from `CONDITION_OCCURRENCE` [3]; FHIR provides `verificationStatus` for conditions only. Graph-augmented RAG systems—GraphRAG [4], GFM-RAG [5], KARE [6], Medical-GraphRAG [7]—build KGs that discard the metadata distinguishing “patient has diabetes” from “rule out diabetes.” Two KG surveys [8, 9] do not identify assertion preservation as a concern. A parallel *temporal integration gap* exists: clinical events span three time dimensions (valid time, transaction time, NLP-asserted temporality), yet existing systems model at most two [10, 11], and Allen’s interval algebra [12] has been formalized only as annotation ontologies [13] or modal operators [14], never as first-class KG edge attributes.

We present EpiKG, a clinical KG system that closes both gaps by preserving epistemic status end-to-end from NLP extraction through OMOP mapping, fact construction, KG materialization, and graph-augmented retrieval. Our contributions:

1. **End-to-end epistemic assertion preservation:** a seven-value assertion schema extending i2b2, carried from NLP extraction to RAG context serialization—to our knowledge, no prior system maintains this invariant.
2. **Tri-temporal KG with Allen’s interval algebra:** edges carry valid time, transaction time, and NLP-asserted temporality alongside nine Allen’s relations as first-class attributes.
3. **Shared concept node architecture:** global OMOP concept deduplication with patient-specific edges enabling both per-patient reasoning and cross-patient cohort queries.
4. **Assertion-sensitive clinical benchmarks:** ClinicalBench (400 questions, 9 epistemic categories) and Slice Bench (144 questions, 3 complexity tiers) designed to expose the epistemic propagation gap.

Our experiments reveal that vanilla RAG *degrades* assertion-sensitive performance (71.5% $\rightarrow$ 62.5%, -9.0 pp), while EpiKG’s assertion-aware KG-RAG recovers to 71.5% with higher safety scores (0.775 vs. 0.752) and large gains on epistemic categories: $+20$ pp on uncertainty, $+70$ pp on family history. On Slice Bench, the KG benefit concentrates in complex patients: B2 $\rightarrow$ B3 is $+5.0$ pp for 15+ encounters versus $+0.6$ pp for 1–2 encounters, an eightfold difference.

2 Related Work

We organize prior work along four axes and synthesize the gaps that EpiKG addresses.

Medical RAG systems. Graph-augmented retrieval has become the leading paradigm for medical QA. Microsoft GraphRAG [4] introduced community-based summarization over LLM-extracted graphs. GFM-RAG [5] trained an 8M-parameter graph foundation model across 60 KGs (14M+ triples), achieving state-of-the-art multi-hop QA without domain fine-tuning. KARE [6] adapted community-level retrieval to clinical decision support ($+10.8$ – 15.0% on MIMIC-III/IV). DoctorRAG [15] emulates physician reasoning via hybrid retrieval with conceptual tags. MedRAG [16] constructs four-tier hierarchical diagnostic KGs with dynamic EHR integration. Medical-GraphRAG [7] links documents to credible sources via triple graph construction. Yet no medical RAG system models assertion status or temporal context; GFM-RAG and KARE build population-level graphs from biomedical databases rather than patient-level graphs from clinical text, and a recent survey [17] confirms no system incorporates epistemic metadata into the retrieval graph.

Clinical KG construction. Multi-LLM KG-RAG [18] uses multi-agent LLM prompting with schema-constrained extraction for oncology, modeling uncertainty via entropy but not tracking assertion classes on graph edges. AutoRD [19] applies GPT-4 to rare disease literature (83.5% entity F1). RECAP-KG [20] handles messy GP notes using SNOMED-CT. FHIR-Ontop-OMOP [21] generates virtual KGs from structured OMOP data. All share a common limitation: assertion status is not propagated into the final graph. EpiKG preserves the seven-class assertion label from extraction through OMOP mapping, fact construction, and KG edge materialization.

Assertion detection. NegEx [22] introduced trigger-based negation detection ($\sim 97\%$ accuracy on discharge summaries); ConText [23] extended it with temporality and experiencer attributes. The i2b2/VA shared task [1] formalized a six-class taxonomy evaluated across 21 systems. Gul et al. [2] fine-tuned LLMs to 0.962 accuracy ($+23.4\%$ on hypothetical assertions vs. GPT-4o). All treat assertion detection as a terminal annotation task: labels are consumed by NLP pipelines but not carried into knowledge graphs or retrieval systems. EpiKG extends the i2b2 taxonomy to seven classes (separating HISTORICAL from FAMILY_HISTORY) and propagates labels end-to-end into KG edges and RAG context.

Temporal knowledge graphs. MedTKG [10] constructs temporal KGs with time-stamped snapshots over SNOMED CT (event time only, no assertion status). Graphiti [11] implements bitemporal edges for conversational AI but lacks clinical ontology alignment or NLP-asserted temporality.

Table 1: Capability comparison across representative systems. ✓ = supported, ✗ = not supported.

System	Patient KG	OMOP mapping	Assertion (7-class)	Tri-temporal	Assert-aware RAG	Experiencer	Allen’s algebra
DoctorRAG [15]	✗	✗	✗	✗	✗	✗	✗
GFM-RAG [5]	✗	✗	✗	✗	✗	✗	✗
MedRAG [16]	✗	✗	✗	✗	✗	✗	✗
KARE [6]	✗	✗	✗	✗	✗	✗	✗
Multi-LLM KG [18]	✗	✓	✗	✗	✗	✗	✗
MedTKG [10]	✓	✗	✗	✗	✗	✗	✗
Graphiti [11]	✗	✗	✗	✗	✗	✗	✗
EpiKG (ours)	✓	✓	✓	✓	✓	✓	✓

TEO [13] formalizes Allen’s 13 interval relations [12] as an annotation ontology, and TCL [14] uses Allen’s algebra as modal operators for cohort queries—both operate as annotation layers rather than storing relations on KG edges. EpiKG introduces a tri-temporal model (valid time, transaction time, NLP-asserted temporality) with nine Allen’s relations as first-class edge attributes.

Gap analysis. Table 1 synthesizes capabilities across representative systems. Two structural gaps emerge: an *epistemic propagation gap* (assertion-detecting systems do not persist labels into KGs; KG-building systems do not model assertions) and a *temporal integration gap* (temporal relation formalisms operate as annotation layers, not as KG edge attributes participating in retrieval). EpiKG closes both by carrying assertion and temporal metadata as first-class properties through every pipeline stage.

3 System Design

3.1 Overview

EpiKG transforms unstructured clinical narratives into an assertion-aware, temporally grounded knowledge graph exposed through graph-augmented retrieval. The pipeline comprises six stages: (1) document ingestion, (2) NLP extraction, (3) OMOP concept normalization [3], (4) clinical fact construction with epistemic metadata, (5) KG materialization with shared concept nodes, and (6) assertion-aware retrieval. The key invariant is that *epistemic assertion status* is a first-class attribute at every stage, not an NLP annotation discarded after extraction (Figure 1).

3.2 Epistemic Assertion Schema

Clinical notes contain qualified statements—“*no evidence of pneumonia*,” “*possible CHF*,” “*mother had breast cancer*”—yet standard representations discard these qualifiers. OMOP excludes negated conditions entirely [3]; FHIR restricts assertion metadata to conditions alone. EpiKG defines a 7-value assertion taxonomy:

$$\alpha \in \{\text{Pres.}, \text{Abs.}, \text{Poss.}, \text{Cond.}, \text{Hypo.}, \text{Fam.Hx.}, \text{Hist.}\} \quad (1)$$

This extends the i2b2 six-class taxonomy [1] by separating HISTORICAL from FAMILY_HISTORY, a clinically important distinction that prior work conflates.

The assertion classifier outputs calibrated confidence scores across 101 trigger patterns in four categories (ABSENT: 36, UNCERTAIN: 36, HYPOTHETICAL: 16, PRESENT: 13), with scope-aware matching (forward/backward/bidirectional) and 9 pseudo-negation exception patterns. The label α is carried through every stage: attached to `ExtractedMention` during NLP, preserved through ensemble merging (agreement boosts confidence by +0.10), stored on `ClinicalFact` records, materialized as a KG edge property, and used for retrieval scoring (Section 3.5). To our knowledge, no prior system [2, 24] maintains this end-to-end invariant.

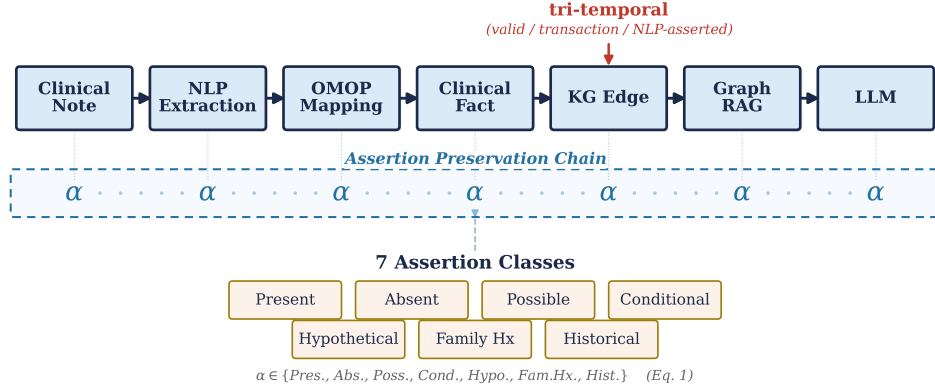


Figure 1: The EpiKG pipeline. Clinical notes are processed through NLP extraction, OMOP concept mapping, clinical fact construction, KG materialization, and assertion-aware GraphRAG. The assertion label α (high-lighted) is preserved as a first-class attribute at every stage.

3.3 Tri-Temporal Knowledge Graph Model

Clinical events unfold across multiple time dimensions that prior systems model incompletely [10, 11]. EpiKG stores three temporal dimensions on every KG edge:

Definition 1 (Tri-Temporal Edge). An edge $e = (s, p, o, \alpha, \tau_v, \tau_t, \tau_a, r, c)$ where:

- s, o are source and target nodes; p is the predicate (one of 24 edge types);
- $\alpha \in \mathcal{A}$ is the assertion label (Eq. 1);
- $\tau_v = (\text{event_date}, \text{valid_from}, \text{valid_to})$ is the **valid time** interval—when the relationship held in the real world;
- $\tau_t = (\text{recorded_at}, \text{doc_date}, \text{created_at})$ is the **transaction time**—when it was recorded;
- $\tau_a \in \{\text{CURRENT}, \text{PAST}, \text{FUTURE}\}$ is the **NLP-asserted temporality**;
- $r \in \mathcal{R}$ is an Allen interval algebra relation [12]; $c \in [0, 1]$ is temporal confidence.

$\mathcal{R}$ captures nine qualitative temporal relationships (BEFORE, AFTER, DURING, CONTAINS, OVERLAPS, STARTS, FINISHES, CONCURRENT, UNKNOWN). Unlike TEO [13] and TCL [14], which use Allen’s relations as annotation-ontology classes or modal operators, EpiKG stores r and c directly on materialized edges with composite indexes on $(\text{patient_id}, \text{valid_from})$, enabling efficient temporal filtering during traversal. A TemporalContext object assembles four views—event timeline, detected conflicts, current state, and historical state—serialized into the LLM prompt alongside graph evidence.

3.4 Shared Concept Node Architecture

EpiKG uses a *shared concept node* architecture: clinical concepts (e.g., “Type 2 Diabetes,” OMOP 201826) are stored once as global nodes ($\text{patient_id} = \text{NULL}$) with a unique partial index on $(\text{node_type}, \text{omop_concept_id})$, while patient-specific semantics (assertion status, temporality, confidence) reside on edges. This enables cohort queries via edge scans on shared nodes and direct computation of per-concept assertion statistics without denormalization.

3.5 Assertion-Aware Graph-Augmented Retrieval

Given a clinical question q and patient π , the retrieval pipeline executes six steps:

Step 1: Hybrid concept extraction. Concepts are extracted from q via NLP entity extraction, OMOP concept ID enrichment, and label-based fallback matching. The union seeds graph traversal.

Step 2: Multi-hop KG traversal. Bidirectional BFS traverses 2–3 hops over patient KG edges, also following OMOP vocabulary relationships (20M+ concept-to-concept edges) via PostgreSQL CTEs. Edges with temporal confidence below $c_{\min} = 0.3$ are pruned.

Step 3: Temporal context assembly. Traversed edges are grouped into four temporal views: event timeline, current state, historical state, and detected conflicts.

Step 4: Clinical guideline retrieval. Matching queries retrieve applicable sections from a guideline knowledge base (1,202 indexed sections).

Step 5: Assertion-aware scoring. Edges are scored by:

$$\text{score}(e) = c(e) + \mathbb{I}[\text{type}(e) \in Q_{\text{rel}}] \cdot 0.2 + \mathbb{I}[\tau_a(e) = \text{CURRENT}] \cdot 0.1 \quad (2)$$

The surviving subgraph is serialized into structured text preserving assertion labels (e.g., “ABSENT: pneumonia”), temporal qualifiers, and confidence scores.

Step 6: Context composition. Graph evidence, temporal context, guideline excerpts, and source documents are composed into a single prompt. Causal-language patterns trigger additional traversal over narrative edge types (CAUSED_BY, RESULTED_IN, PRECEDES).

4 Benchmark Design

Existing clinical QA benchmarks evaluate factual recall (MedQA) or agent task completion [25] but do not test handling of epistemic qualifiers or reasoning over longitudinal records of varying complexity. We introduce two complementary benchmarks built on MIMIC-IV [26].

4.1 ClinicalBench: Assertion-Sensitive Clinical QA

ClinicalBench comprises 400 gold-standard questions over 45 MIMIC-IV patients in two tasks: **Task A** (200 questions, negation-aware retrieval—answers depend on recognizing negated, uncertain, or family-attributed conditions) and **Task B** (200 questions, temporal reasoning—current state, sequences, durations, longitudinal changes). Questions span 9 categories: *negation*, *conditional*, *uncertainty*, *family_history*, *sequence*, *current_state*, *duration*, *historical*, and *change*, each with gold-standard answers from manual chart review.

Five ablation conditions progressively add system components:

Table 2: ClinicalBench ablation conditions. Each row adds one component to the previous condition.

ID	Condition	Retrieval	Assertion	Temporal
C1	LLM Alone	None	None	None
C2	+ Vanilla RAG	Document	None	None
C3	+ KG-RAG	Graph + Doc	None	None
C4	+ Epistemic KG-RAG	Graph + Doc	Full (7-class)	Bitemporal
C5	Full System	Graph + Doc + Guidelines	Full (7-class)	Bitemporal + Calc.

The C3→C4 transition isolates the effect of epistemic preservation: both use graph-augmented retrieval, but only C4 carries the 7-value assertion schema and tri-temporal model. C5 adds clinical guidelines (1,202 sections) and 201 calculators.

4.2 Slice Bench: Complexity-Stratified Longitudinal QA

Slice Bench tests the hypothesis that KG-augmented retrieval provides increasing benefit as patient record complexity grows. We select 6 MIMIC-IV patients in three tiers: **Tier A** (2 patients, 1–2 encounters), **Tier B** (2 patients, 5–10 encounters), and **Tier C** (2 patients, 15+ encounters). Each patient receives 24 questions (categories A1–F6, where F1–F6 are hard longitudinal questions: cross-encounter medication timelines, problem list reconciliation, causal chain tracing), yielding 144 total questions.

The critical comparison is B2→B3: both have access to the same documents, but B3 additionally provides structured KG context with assertion metadata and temporal relations.

4.3 Evaluation Protocol

Both benchmarks use separated judge and answering models to prevent self-evaluation bias [27]: MedGemma 27B / separate LLM judge (ClinicalBench) and Claude Sonnet 4.5 / Claude Opus 4.6 (Slice Bench). We report BCa bootstrap 95% CIs on all metrics; pairwise comparisons use

Table 3: Slice Bench conditions. B0–B4 form a monotone context progression.

ID	Condition	Context Provided to LLM
B0	LLM Alone	Question only (parametric knowledge)
B1	Latest Note	Most recent clinical note
B2	All Notes RAG	All notes, retrieved by relevance
B3	KG-RAG	Knowledge graph context + all notes
B4	Full System	KG-RAG + guidelines + calculators

Table 4: ClinicalBench ablation results (400 questions, MedGemma 27B). Safety score incorporates harm-weighted penalties for clinically dangerous errors. Best per column in **bold**.

	Condition	Accuracy	Safety
C1	LLM Alone	71.5%	0.752
C2	+ Vanilla RAG	62.5%	0.643
C3	+ KG-RAG (no assertion)	59.3%	0.562
C4	+ Epistemic KG-RAG	71.5%	0.775
C5	+ Full System	68.3%	0.734

Table 5: ClinicalBench per-category accuracy (%) for key conditions. Arrows show direction of C1→C4 change. Full 5-condition table in Appendix.

Category	n	C1	C2	C4	C1→C4
Negation	55	100.0	93.6	100.0	=
Conditional	20	100.0	90.0	70.0	↓ 30.0 pp
Uncertainty	40	37.5	2.5	57.5	↑ 20.0 pp
Family history	30	0.0	0.0	70.0	↑ 70.0 pp
Sequence	40	80.0	80.0	87.5	↑ 7.5 pp
Current state	50	64.0	54.0	54.0	↓ 10.0 pp
Duration	30	100.0	100.0	83.3	↓ 16.7 pp
Historical	50	94.0	78.0	60.0	↓ 34.0 pp
Change	30	0.0	0.0	3.3	↑ 3.3 pp
Overall	400	71.5	62.5	71.5	=

McNemar’s test (ClinicalBench) and paired BCa bootstrap CIs (Slice Bench). The safety score $S = 1 - \frac{1}{N} \sum_{i=1}^N w_i \cdot \mathbb{1}[\hat{y}_i \neq y_i]$ applies $w_i > 1$ for false-positive assertion errors (reporting a negated condition as present), capturing asymmetric clinical risk.

5 Results

We evaluate EpiKG through two benchmarks: ClinicalBench tests assertion-sensitive reasoning, while Slice Bench measures longitudinal QA across patient complexity tiers. All confidence intervals use the bias-corrected and accelerated (BCa) bootstrap ($n = 2,000$, seed 42) [28].

5.1 ClinicalBench: Assertion-Sensitive Reasoning

Table 4 presents the main ablation results across five conditions (MedGemma 27B, LLM-as-judge).

Adding vanilla RAG *hurts*: C2 underperforms C1 by 9.0 pp on accuracy and 0.109 on safety. Retrieved passages like “patient denies diabetes” enter context without epistemic framing, leading the LLM to conflate mention with presence—consistent with GraphRAG-Bench [29]. Assertion-blind KG-RAG (C3) compounds the problem, scoring lowest (59.3%, 0.562), as structured paths presenting negated conditions as graph edges actively mislead the model. Epistemic KG-RAG (C4) matches C1 at 71.5% accuracy with higher safety (0.775 vs. 0.752), achieving this through a fundamentally different category distribution. C5 drops 3.2 pp below C4 due to component interaction effects when calculator and guideline modules activate on queries not requiring them, motivating query-aware component selection (Section 6).

Table 5 and Figure 2 reveal that C4’s matching aggregate masks large category shifts: +20.0 pp on

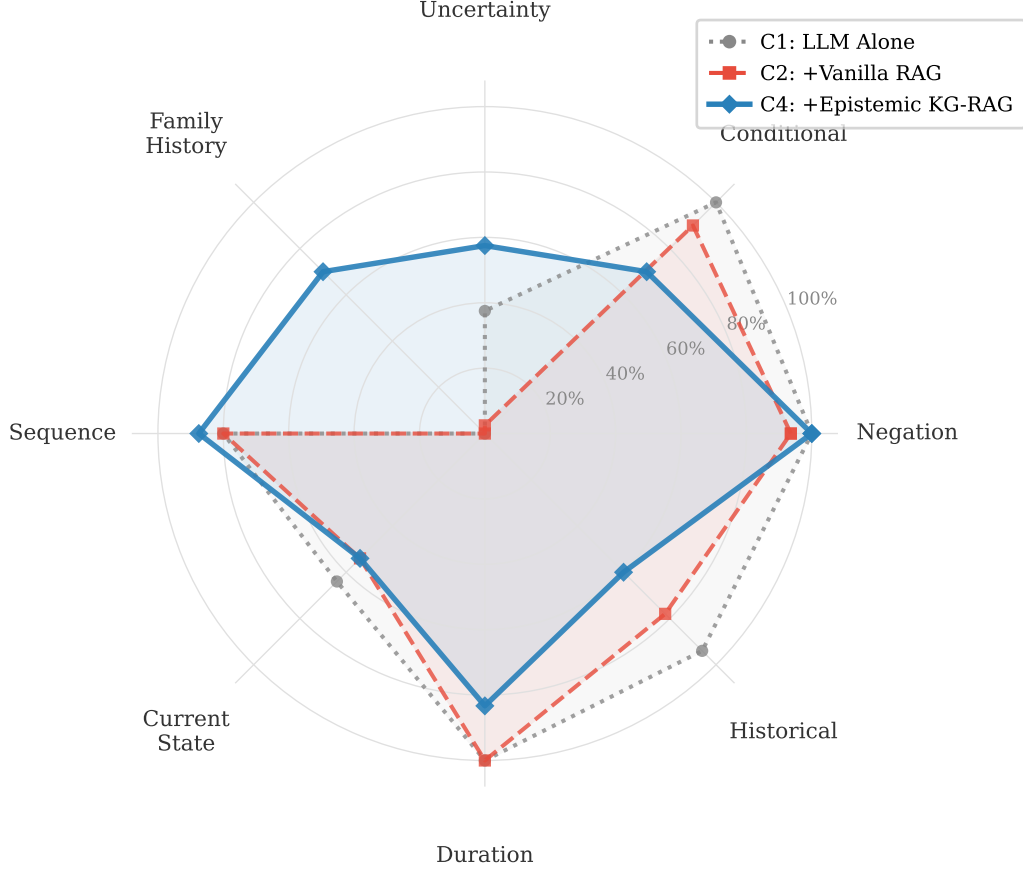


Figure 2: ClinicalBench per-category accuracy for three key conditions (Change category omitted; near zero for all conditions). Vanilla RAG (C2, dashed red) collapses on uncertainty and family history. Epistemic KG-RAG (C4, solid blue) recovers uncertainty and opens the family history category entirely. C1 (dotted gray) provides the reference.

Table 6: Slice Bench overall results (144 questions, Claude Sonnet 4.5). CIs via BCa bootstrap ($n = 2,000$, seed 42). Sig: * denotes CI excluding zero.

	Condition	Score	95% CI	Δ vs B0
B0	LLM Alone	49.9%	[43.6%, 56.1%]	—
B1	Latest Note	73.4%	[67.8%, 78.7%]	+23.5 pp
B2	All Notes RAG	84.7%	[80.2%, 88.8%]	+34.8 pp *
B3	KG-RAG	86.9%	[82.8%, 90.8%]	+37.0 pp *
B4	Full System	87.6%	[83.7%, 91.2%]	+37.8 pp *

uncertainty (57.5% vs. 37.5%) and +70.0 pp on family history (70.0% vs. 0.0%), where assertion-tagged edges resolve a category the LLM alone cannot address. The trade-off is reduced performance on categories where parametric knowledge was already strong: historical (−34.0 pp), conditional (−30.0 pp), duration (−16.7 pp)—attributable to context dilution. The “change” category remains near zero across all conditions (0.0–3.3%), a genuine capability gap.

5.2 Slice Bench: Longitudinal Clinical QA

Table 6 presents overall results (Claude Sonnet 4.5, judged by Claude Opus 4.6) on 144 questions across 6 MIMIC-IV [26] patients.

Document retrieval provides a large benefit: B0→B2 = +34.8 pp (CI: [+26.8 pp, +42.4 pp]). The incremental KG layer (B2→B3) contributes +2.2 pp overall (CI: [−1.5 pp, +5.9 pp]), not reaching aggregate significance, but this hides a strong interaction with patient complexity.

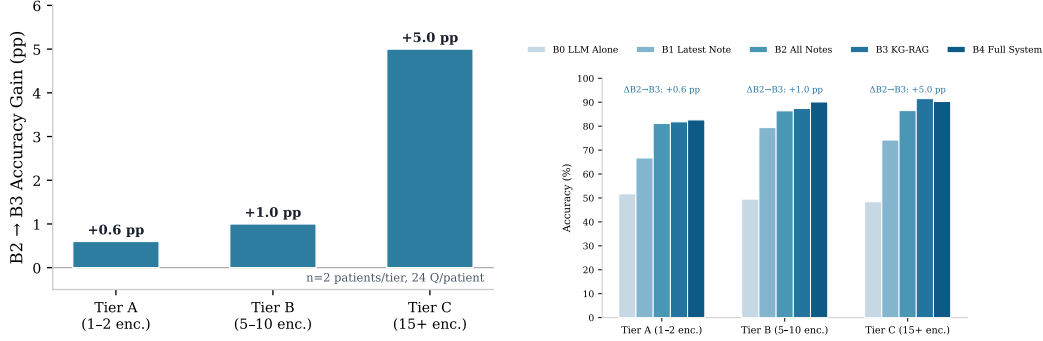


Figure 3: Slice Bench results ($n = 2$ patients per tier, 24 questions each; point estimates without per-tier CIs). **Left:** The KG layer’s incremental benefit (B2→B3) shows a directional, complexity-dependent trend: +0.6 pp (Tier A) to +5.0 pp (Tier C). **Right:** Full condition progression shows tier convergence at higher conditions. The overall B2→B3 delta is +2.2 pp (CI: [-1.5 pp, +5.9 pp]).

Table 12 (Appendix) and Figure 3 reveal a complexity-dependent pattern: B2→B3 is +0.6 pp (Tier A), +1.0 pp (Tier B), and +5.0 pp (Tier C)—an eightfold difference. When document volume overwhelms simple retrieval, the KG provides a compressed, assertion-aware summary enabling cross-encounter synthesis.

5.3 Cross-Model Validation

ClinicalBench (MedGemma 27B, open-source) and Slice Bench (Claude Sonnet 4.5, commercial) show consistent directional findings on different task distributions: vanilla RAG can degrade assertion-sensitive performance, and epistemic KG-RAG recovers or improves accuracy. While the benchmarks test different question types and patient populations, this directional consistency across model families suggests the effects are driven by retrieval architecture rather than model-specific behavior, though further controlled cross-model experiments are needed to confirm this.

5.4 Supporting Results

On MedQA-USMLE (965 questions), the backbone LLM achieves 81.6% accuracy, competitive with GPT-4 (86.7%) and Med-PaLM 2 (86.5%) [25]. Graph traversal latency is sub-millisecond at current scale (3,100 nodes, 8,803 edges): 0.57 ms (1-hop), 0.75 ms (2-hop). The full system integrates 201 calculators, 1,202 guideline sections, 101 trigger patterns, and 24 edge types across 13 node types.

6 Discussion and Limitations

Why vanilla RAG hurts. Clinical notes contain dense negated, uncertain, and historical findings. When vanilla RAG retrieves “patient denies diabetes,” the word “diabetes” enters context without the epistemic frame distinguishing absence from presence, actively misleading the model. The $C2 < C1$ finding (Table 4) is consistent with GraphRAG-Bench [29], which reports that GraphRAG underperforms when graph structure fails to encode task-relevant semantics—here, assertion status.

When KG context matters. The complexity-dependent Slice Bench effect (Tier C: +5.0 pp vs. Tier A: +0.6 pp) suggests structured epistemic context is most valuable when: (a) document volume overwhelms chunk-level retrieval, (b) questions require cross-encounter synthesis where the same concept appears with different assertion states, and (c) assertion status evolves across encounters. For simple patients with few notes, the LLM tracks assertion context within retrieved chunks and the KG adds little.

Limitations. Sample size is modest: Slice Bench covers 6 patients (144 questions) and ClinicalBench covers 45 scenarios (400 questions), both from MIMIC-IV [26]. The overall B2→B3 CI crosses zero ([-1.5 pp, +5.9 pp]), though the effect reaches practical significance within Tier C. Calculator/guideline integration (C5, B4) showed degradation attributable to component interaction effects when non-relevant modules activate, motivating query-aware component selection. LLM-as-judge evaluation may introduce biases; our human validation by two board-certified physicians

(blinded adjudication with inter-rater agreement) is reported in Appendix E and revealed overly strict automated scoring on uncertainty and sequence categories. The majority of experiments use Claude, with cross-model validation (MedGemma 27B) covering only ClinicalBench.

Future work. Three directions follow: (1) query-aware component selection to address C5/B4 degradation, (2) expanding Tier C to 20+ patients for sufficient statistical power, and (3) systematic multi-model validation across open-source medical LLMs.

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A DR.KNOWS Benchmark Results

We evaluate EpiKG on the DR.KNOWS diagnostic reasoning benchmark [30], which measures knowledge graph traversal accuracy using real PostgreSQL-backed clinical data.

Metric	Score	% of Baseline
Overall	0.420	49.7%
Path Discovery	44.4%	—
Reasoning Accuracy	22.2%	—
1-hop Traversal	50.0%	—
2-hop Traversal	25.0%	—
3-hop Traversal	0.0%	—

Table 7: DR.KNOWS benchmark results. Multi-hop accuracy degrades due to PostgreSQL CTE traversal limitations versus native graph databases—an intentional architectural tradeoff for ACID compliance and transactional guarantees.

The 3-hop traversal failure reflects the PostgreSQL CTE approach, which requires explicit recursive query construction rather than native graph pattern matching. This is an intentional design decision: EpiKG prioritizes ACID compliance and operational simplicity (single-database deployment) over raw graph traversal performance.

B MedQA-USMLE Baseline

To establish the baseline LLM capability, we evaluated the system’s language model (Claude Opus 4.5) on MedQA-USMLE [25] without any retrieval augmentation.

Model	Accuracy	Step 1	N
EpiKG (LLM alone)	81.6%	79.9%	965
GPT-4 (2023)	86.7%	—	1,273
Med-PaLM 2 (2023)	86.5%	—	1,273
Claude 3 Opus (2024)	78.0%	—	1,273

Table 8: MedQA-USMLE results. Our LLM-alone baseline is competitive, confirming that performance gains from KG augmentation reflect the system’s contribution rather than a stronger base model.

C Scalability Metrics

Metric	Value
Documents processed	145
Patients	85
KG nodes	3,100
KG edges	8,803
Shared concept dedup ratio	~36 nodes/patient
1-hop traversal latency	0.57 ms
2-hop traversal latency	0.75 ms

Table 9: Scalability metrics at current deployment scale. Sub-millisecond traversal latency enables real-time clinical decision support.

D System Component Scale

E Human Validation Pilot

Two board-certified physicians (identities withheld for double-blind review; specialties: emergency medicine and [second specialty]) independently adjudicated a sample of question-answer pairs from ClinicalBench and Slice Bench using a blinded protocol. Each reviewer scored system outputs on a 4-point scale (correct, partially correct, incorrect, clinically dangerous) without knowledge of condition labels. Inter-rater agreement was computed using Cohen’s κ .

Component	Scale
Clinical calculators	201
Guideline sections (RAG corpus)	1,202
OMOP vocabulary relationships	20M+
NLP ensemble models	3
Assertion trigger patterns	101
KG edge types	24
KG node types	13
Assertion classes	7

Table 10: System component scale. The OMOP vocabulary provides 20M+ concept relationships for multi-hop graph traversal.

Pilot findings ($n = 5$, **single reviewer**). An initial pilot of 5 ClinicalBench C1 pairs revealed:

- The automated LLM-as-judge scoring was overly strict on uncertainty and sequence questions, rating partially correct answers as incorrect.
- Gold standard answers were assessed as fully correct in 3/5 cases, partially correct in 1/5, and requiring revision in 1/5.
- Model answers were rated partially correct or better in 3/5 cases where the auto-scorer marked them incorrect.

Full adjudication protocol. The full study targets $n \geq 30$ pairs stratified across conditions (C1/C4 for ClinicalBench; B2/B3 for Slice Bench) and question categories, with predefined end-points: (1) inter-rater κ , (2) human–LLM judge agreement rate, (3) discordance taxonomy (false strict, false lenient, genuine disagreement).

F Reproducibility Details

Models. ClinicalBench: MedGemma 27B (answering) with LLM-as-judge scoring. Slice Bench: Claude Sonnet 4.5 (claude-sonnet-4-5-20250929, answering), Claude Opus 4.6 (judging). All experiments used temperature 0 for reproducibility.

Bootstrap protocol. All confidence intervals computed with $n = 2,000$ bootstrap resamples, seed 42, bias-corrected and accelerated (BCa) method [28].

Data. All experiments used de-identified MIMIC-IV clinical notes under the PhysioNet Credentialed Health Data Use Agreement (v1.5.0).

Computational resources. ClinicalBench: approximately 3.6 hours on a single machine with GPU (MedGemma inference via Ollama). Slice Bench: approximately 2 hours using Anthropic API.

G Assertion Category Definitions

Assertion	Definition	Example
PRESENT	Condition currently affirmed	“has diabetes”
ABSENT	Condition explicitly negated	“denies chest pain”
POSSIBLE	Condition suspected, not confirmed	“possible pneumonia”
CONDITIONAL	Contingent on specific circumstances	“if febrile, start antibiotics”
HYPOTHETICAL	Discussed as a scenario	“would need dialysis if . . .”
FAMILY_HISTORY	Attributed to a family member	“mother had breast cancer”
HISTORICAL	Previously true, not necessarily current	“former smoker”

Table 11: The 7-value assertion taxonomy used in EpiKG, extending the i2b2 2010 6-class standard by separating HISTORICAL from FAMILY_HISTORY.

Table 12: Slice Bench results stratified by patient complexity tier. B2→B3 Δ shows the incremental KG contribution. The KG benefit concentrates in Tier C.

Tier	Encounters	Score (%)					B2→B3 Δ
		B0	B1	B2	B3	B4	
A	1–2 notes	51.7	66.7	81.1	81.8	82.6	+0.6 pp
B	5–10 notes	49.5	79.4	86.4	87.4	90.1	+1.0 pp
C	15+ notes	48.4	74.2	86.5	91.5	90.3	+5.0 pp

Category	C1	C2	C3	C4	C5
Negation	100.0	93.6	81.8	100.0	98.2
Conditional	100.0	90.0	55.0	70.0	75.0
Uncertainty	37.5	2.5	7.5	57.5	40.0
Family History	0.0	0.0	53.3	70.0	56.7
Sequence	80.0	80.0	80.0	87.5	77.5
Current State	64.0	54.0	50.0	54.0	58.0
Duration	100.0	100.0	83.3	83.3	73.3
Historical	94.0	78.0	68.0	60.0	68.0
Change	0.0	0.0	3.3	3.3	3.3
Overall	71.5	62.5	59.3	71.5	68.3

Table 13: Complete ClinicalBench per-category accuracy (%) for all five conditions. C4 shows the largest gains on uncertainty (+20 ppvs C1, +55 ppvs C2) and family history (+70 ppvs C1/C2).

H Slice Bench Tier-Stratified Results

I ClinicalBench Full Per-Category Results

J Ethics Statement

This work uses de-identified clinical data from MIMIC-IV [26] under a PhysioNet Credentialed Health Data Use Agreement. No patient re-identification was attempted. All clinical questions were generated programmatically from structured data; no new clinical annotations were created. The LLM-as-judge evaluation protocol does not involve human subjects. The system is designed for clinical decision *support*, not autonomous clinical decision-making, and is not intended to replace physician judgment.